



Society of Petroleum Engineers

SPE-229604-MS

The Integrated Methodology for Improved Drilling Performance in Ultra-Hard and Highly Heterogeneous Shale formation

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229604-MS](https://doi.org/10.2118/229604-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

With the breakthrough in shale oil exploration in the Fengcheng formation, this unconventional resource has become the main potential contributor of successive hydrocarbon for the Mahu, Xinjiang Oil Field. However, this ultra-hard and highly heterogeneous shale formation, with CCS ranging from 190 to 300 MPa, low ROP, short drilling footage, frequent concentric or grooving drill bit wear, and a long drilling cycle during drilling the third interval of horizontal wells, posed a great challenge to the cost-effective development of this newly tapped resource.

The integrated methodology has been developed to address the issues posed by this unique geological challenge. Three-dimensional drillability prediction for pre-drill formation has been achieved through the innovative development of an unsupervised neural network clustering method for drillability prediction, based on modelling that combines seismic, logging and real-time resistivity data. This method enables both ROP and formation drillability to be predicted along the wellbore trajectory, helping to specify customized optimization solutions for the trajectory and drill string design of vertical, curved and horizontal sections. Indoor drilling parameter sensitivity tests based on sampled rock cores indicated that the WOB threshold for the Fengcheng formation should exceed 100 kN, with increasing WOB serving as the primary stimulus for enhancing ROP compared to RPM. Inverted bottom hole assembly designs, including a modified down-hole motor, RSS and HWDP, which are applicable for adopting a high WOB, have been adopted to increase ROP during horizontal drilling. Furthermore, highly efficient drill bits have also been developed and customized based on drillability prediction results along the wellbore trajectory. These bits include both PDC and impregnated diamond bits with high impact and wear resistance characteristics.

The vertical and horizontal distribution characteristics of the drillability of the Fengcheng formation could be clarified based on the developed prediction method, which categorized the sections into four groups based on actual resistivity and drilling ROP data. Curved sections in the trajectory have been

deliberately designed to avoid ultra-hard formations. Drill string designs, including a 5-blade PDC bit + motor + MWD, a 6-blade PDC bit + RSS + motor, and an impregnated diamond bit + steerable turbine + LWD/MWD, have been tailored for sections with varying drillability. Using high WOB with the proposed optimized drill string designs increased ROP by 20–50%, with WOB increasing from 80–120 kN to 160–200 kN. Furthermore, customized bit design criteria were formulated for vertical, curved and horizontal sections, with the drilling footage increased by 12% to 200% and the ROP by more than 20% respectively. Improved drilling performance was observed continuously during four rounds of horizontal wells with the implementation of this developed integrated methodology. The average drilling cycle was shortened from 119 days to 77 days with average ROP being increased from 4.4m/h to 7.2m/h.

By summarizing the challenge, methodology and achievement during horizontal well drilling in this ultra-hard and highly heterogeneous shale formation, this paper provides a valuable reference for the cost-effective exploration and development of the similar unconventional hydrocarbon resource.

Introduction

As the exploration and production of global hydrocarbon resources moves into deeper waters and unconventional reservoirs, unconventional oil and gas resources, including shale oil and gas as well as tight oil and gas, have also become important successors to China's hydrocarbon reserves and production in the future. According to statistics, the annual production of shale and tight gas and oil amounts to 80 billion cubic meters and 17 million tons respectively, accounting for nearly 39% and 10% of total domestic production. Moreover, it is estimated that total unconventional hydrocarbon production will account for half of national output in the future (Jia.,2024). Most of China's shale oil is buried in the Junggar Basin, the Ordos Basin, the Bohai Bay Basin, the Sichuan Basin and the Songliao Basin (Wang et al.,2024). The Fengcheng formation in the Mabei block belongs to the Carboniferous period and is another newly discovered shale oil deposit in the Junggar Basin, following the successful development of the Jimsar shale oil deposit (Yi et al., 2024).

The Mabei block is located in the central depression of the Junggar Basin. The Fengcheng formation, which has a total thickness ranging from 200 to 800 meters and is subdivided into three sub-layers P1f1, P1f2 and P1f3, has a burial depth of greater than 4500 meters. The tested confined compression strength (CCS) of Fengcheng formation rock ranges from 190 to 300 MPa, which is equal to or greater than that of granite. Despite its burial depth and the fact that it is extremely difficult to drill, the formation is brittle due to a high difference in geo-stress and its mineral content, which mainly consists of carbonate, quartzose and cloudy shale. Furthermore, the formation's high heterogeneity creates favorable conditions for the expansion and connection of fractured fractures. The three-spud design of horizontal wells and hydraulic fracturing have been adopted for the development of this newly discovered shale oil resource. The vertical, curved and horizontal sections of the third drilling interval, which amount to a total length of 1600–2800 meters, are mainly concentrated in the Fengcheng formation. Except for the risk of gas intrusion and wellbore leakage due to developed fractured zones, ultra-hard and highly heterogeneous formation have served as the predominant challenge for shortening drilling cycle and costs due to low ROP, short drilling footage, multiple drilling trips and severe drill bit wear while drilling the third interval. The drilling cycle for the pilot horizontal wells in the area reached nearly 160 days with more than 100 of those days being spent on third-interval drilling alone, restricting the cost-effective development of this newly tapped unconventional resource.

This paper presents the challenges encountered while drilling in this ultra-hard and highly heterogeneous shale formation, then the integrated methodology, developed for increasing ROP as well as drilling footage, reducing drilling trips, alleviating drill bit wear and shorten drilling cycle, has also been illustrated for improved drilling performance. The application and implementation of these developed measures achieved significant results as the drilling cycle has been shortened to 77 days with average ROP being increased to 7.2 m/h after the iterative upgrading of drilling technology. These challenge, methodology

and achievement provide the valuable reference for the cost-effective exploration and development of the similar unconventional hydrocarbon resource globally.

Challenges while drilling in the Fengcheng formation

Ultra-hard formation

PDC bits were used to drill the third interval of the pilot horizontal wells in the area. Previously, these sections had taken between 9 and even more than 20 drilling trips to complete, with most drill bits having drilling footage and an average ROP of 30-100 meters and 1 m/h, respectively. Concentric wear and grooving of the drill bit had been observed frequently and consecutively, especially when drilling the curved and horizontal sections, as shown in Fig 1.



Figure 1—frequent and even consecutive concentric or grooving wear for PDC bits

Indoor rock strength tests were conducted on core samples, revealing that the confined compression strength (CCS) of Fengcheng formation rock ranges from 190 to 300 MPa shown in Fig 2. This ultra-hard characteristic justifies low ROP and short drilling footage when drilling in the formation with PDC bits. This also facilitated the development of high-performance drill bits which maintain good performance in both ROP and footage, setting requirements for cutters, design of bit structures and etc.

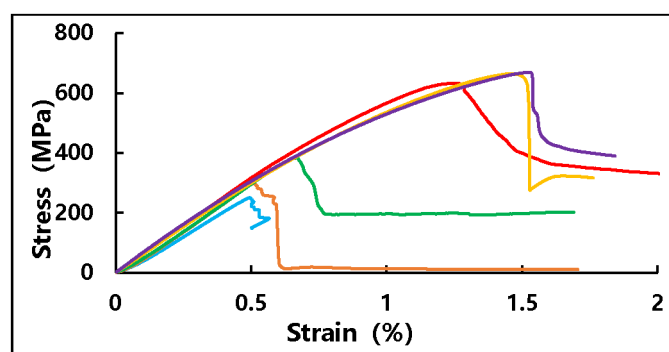


Figure 2—Typical stress-strain curve of rock core sample from Fengcheng formation

Highly heterogeneous in formation drillability

The formation strength's highly heterogeneous characteristics caused significant difference in the drilling footage and ROP when drilling the curved and horizontal sections. The drilling footage and the average ROP for sections of the pilot wells drilled using the same PDC bit were recorded as ranging from 24 to 294 meters and 0.7 to 2.8 m/h, respectively. Severe vibration and shock were recorded down-hole, which also served as another primary cause of short drilling footage, low ROP and severe bit wear. Severe vibration could also lead to a significant increase in the risk of down-hole tool and even drill string failure. It is easy to understand why drilling with a impregnated diamond bit and a turbine in an ultra-hard or highly

abrasive formation would be a viable solution. However, the formation's highly heterogeneous nature makes it difficult to design the optimal drill string combination considering the drilling footage as well as the ROP of PDC and impregnated diamond bits during the pre-drill design phase.

Drilling with large WOB

Indoor tests conducted using a mini PDC bit over ultra-hard rock core samples from the Fengcheng formation, as shown in Fig 3, revealed the sensitivity of ROP to drilling parameters. The test results revealed that the ROP didn't change significantly with the change in RPM but increased exponentially with increase in WOB, as shown in fig 4.



Figure 3—minin-rock drillability testing device (left) and core (right)

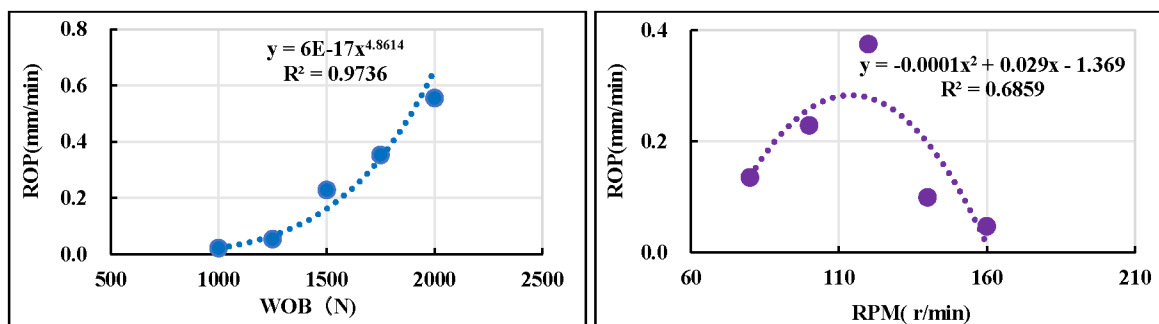


Figure 4—sensitivity test results of ROP to change in WOB and RPM

Furthermore, the test results indicated that the WOB threshold for the PDC bit in an 8.5-inch borehole while drilling in the Fengcheng formation should be no less than 100 kN through proportional conversion. Drill string buckling could easily be induced during horizontal drilling with a high WOB. The components of the BHA, including the RSS, LWD/MWD and motor, were also prone to failure during prolonged drilling with high WOB.

Methodology for improved drilling performance

The integrated methodology, including 3D formation drillability prediction, development of tailored high performance drill bits and technical measures for drilling with high WOB, has been developed to address the challenges posed by drilling in this ultra-hard and highly heterogeneous formation.

Drillability prediction and drill design optimization

Correlation analysis revealed a strong correlation between formation resistivity and ROP during drilling of the curved and horizontal sections of the pilot wells in the area. The drillability of the Fengcheng formation was then classified into four categories based on the resistivity, using the unsupervised neural network cluster analysis method shown as fig 5.

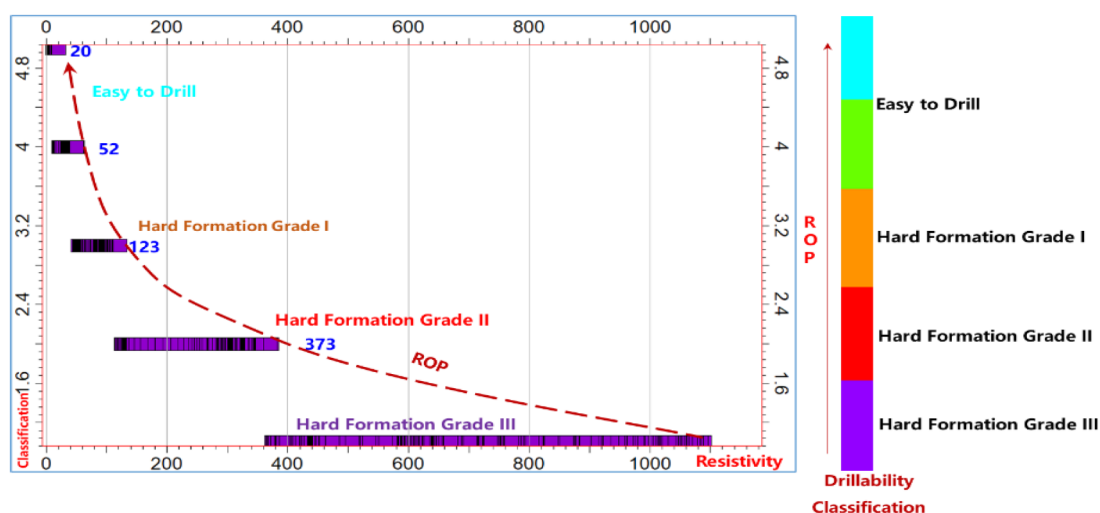


Figure 5—formation drillability classification corresponding to different resistivity thresholds

A three-dimensional formation resistivity prediction model was established which combines real-time or logging resistivity, tectonic level interpretation and seismic data. This model is based on geological statistics and the seismic amplitude phase control method. Three-dimensional drillability prediction could be achieved by applying classification thresholds between formation resistivity and drilling ROP, as defined for the prediction of drillability for a single well. The horizontal and vertical distribution characteristics of the drillability of the Fengcheng formation could be predicted during the pre-drill design phase based on the established model. Most importantly, formation drillability could be predicted along the designed wellbore trajectory, as shown in Fig 6.

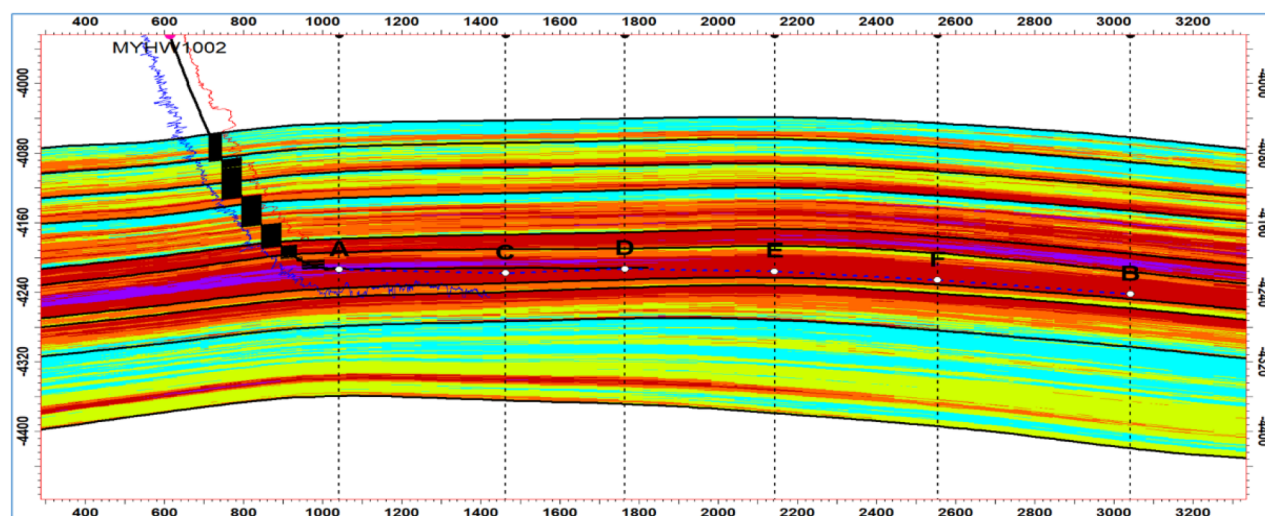


Figure 6—formation drillability prediction along wellbore trajectory

This could help to optimize the trajectory and drill string designs according to the prediction results. Typical optimization could be exemplified by changing the original double $5.6^{\circ}/30\text{m}$ build up curved design into a $6.3^{\circ}/30\text{m}+4.8^{\circ}/30\text{m}$ design in order to increase the dogleg and the length of upper, easy-to-drill curved section. This also increased the length of the lower, hard-to-drill inclined section. The design of the drill string could also be tailored, especially the choice between PDC and impregnated diamond bits, taking into account the ROP, drilling hours and footage of these two options according to the predicted results. In the X-YH well, the formation was predicted to be a hard Grade III formation from 5100 m. The

drill string was then changed to a steerable turbine+impregnated drill bit, replacing the original PDC bit + motor design. Drilling with the impregnated diamond bit yielded a footage of 373 meters, nearly 3.7 times that achieved with PDC bits in the area. More importantly, template for drill string designs had been established based on formation drillability prediction results. For the upper, easy-to-drill, curved section with resistivity lower than $200\Omega\cdot\text{m}$, a five-blade PDC bit with double-row cutters, RSS and a motor was recommended to ensure a high ROP. Meanwhile, an impregnated diamond bit +steerable turbine+LWD/MWD design was recommended for the lower, hard-to-drill, curved section with resistivity greater than $200\Omega\cdot\text{m}$. For horizontal sections, five-blade PDC bit with double-row cutters+RSS+ motor, six-blade PDC bit with double-row cutters+RSS+ motor and impregnated diamond bit +steerable turbine+LWD/MWD design were recommended for formation with resistivity lower than $500\Omega\cdot\text{m}$, $500\Omega\cdot\text{m}$ - $1000\Omega\cdot\text{m}$ and higher than $1000\Omega\cdot\text{m}$.

Development and application of customized high-performance drill bits

Given the high strength of the formation rock and the severe down-hole vibration experienced during drilling in the Fengcheng formation, it is essential to increase the abrasion and impact resistance of PDC cutters in order to increase drilling footage and ROP. The anti-abrasive and impact-resistant performance of different PDC cutters, made using two different machining processes (laser and EDM), with two different depths of cobalt removal and three different formulas, had been investigated to determine the optimal combination that balanced the anti-abrasion and impact resistant characteristics. Non-planar cutters, namely convex-ridge teeth, were selected as the main cutters for PDC bits due to its superior impact resistance. To address frequent and even consecutive concentric or grooving bit wear, conical and axe-shaped teeth were arranged around the centre of the drill bit in order to control the depth of cut (DOC), improve impact resistance and reduce the tendency for bit whirl. Indoor experiments had also been conducted on core samples to study the rock-breaking performance of PDC cutters of different sizes and with different back rakes as well as different elevation difference between the front and rear cutters. The results of these experiments revealed that the optimal specifications for the size, back rake and elevation difference of the proposed cutter were $\phi 13/16$ mm, $10\text{--}15^\circ$ and 1-1.5mm, respectively. In view of the necessity and possibility of slide drilling during curved and horizontal sections, high-performance impregnated diamond bits that possess a good balance between aggressiveness and directional controllability have also been developed to improve drilling performance while drilling the ultra-hard sections.

Based on predictions of the vertical and horizontal distribution characteristics of formation drillability, a series of customized, high-performance drill bits has been developed for the third interval during the Fengcheng formation. This ensures a high ROP and long drilling footage, consequently reducing the number of drilling trips. The five-blade and single-row cutter PDC bit shown as [fig 7\(a\)](#), applicable for soft to medium hard formation with strong aggressiveness, was designed specifically for increasing ROP while drilling the vertical section of the third interval. The six-blade and double-row cutter PDC bits shown as [fig 7\(b\)](#) and [7\(c\)](#), with convex ridge teeth as primary cutters and conical or axe-shaped teeth arranged around drill bit center, were designed to prevent concentric or grooving bit wear with superior impact resistance. These two PDC bits have been primarily used for wellbore sections with a hard formation grade of I to II and applied with an RSS and a down-hole motor during curved and horizontal drilling. For recommended impregnated diamond bit shown as [fig 7\(d\)](#), which has been generally combined with high-speed steerable turbine to drill the ultra-hard formation with a grade of III.

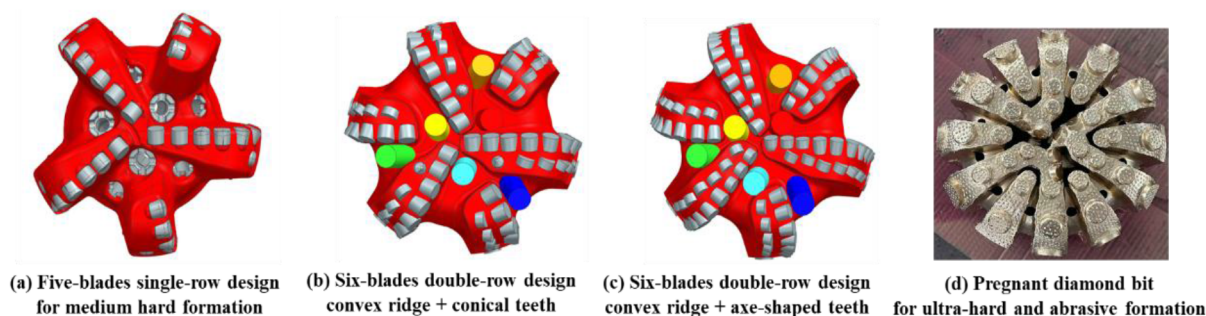


Figure 7—Customized high-efficiency drill bits for different wellbore sections with varied drillability

The recommended PDC bit 7(b) completed the curved section of well Y-2XM in a single drilling trip, achieving a drilling footage of 907 meters and an average ROP of 8m/h, respectively. These two figures set new records for curved section drilling in the area. The recommended PDC bit 7(c) achieved a drilling footage of 489 meters and an average ROP of 3.9m/h while drilling the horizontal section of well FC-6. This total drilling footage set a new record for horizontal drilling in the P_1f_2 sub-layer of the area. Moreover, the drilling footage and the average ROP were increased by 12%~200% and by over 20%, respectively, when recommended drill bits were used for third interval drilling of horizontal wells, compared to PDC bits used for the previous pilot wells according to the recorded data.

Drilling with large WOB

The specific design of the drill string used in the different sections of the third interval could be refined based on the recommended drill bits. For PDC bits, drilling with a WOB up to 150 kN and even larger was beneficial for increasing ROP based on the indoor drilling parameters sensitivity tests. However, the maximum WOB that could be sustained for previously adopted down-hole RSS or motor was no greater than 140 kN. Upgrades had been made to the RSS and motor to accommodate a large WOB. Moreover, the theoretical analysis, over the buckling characteristics of several different inverted drill string designs, was carried out to determine the optimal design for exerting WOB larger than 200 kN without buckling during horizontal drilling. A new drill string design was recommended and adopted for horizontal drilling with large WOB. This high-stiffness design used 5-1/2 inch heavy weight drill pipe (HWDP) to replace the previous 5-1/2 inch drill pipe for the horizontal section.

Pilot horizontal wells were drilled using a recommended PDC bit (7b), a high-stiffness drill string design with upgraded motor and RSS, with a WOB ranging from 160-200 kN. This achieved an average ROP increase between 20% to 50%, compared to previous drilling with a WOB ranging from 80-120 kN. Despite the decrease in footage when drilling with a large WOB in the Fengcheng formation, analysis of the average ROP per round trip demonstrated the advantage of adopting a large WOB for shortening the drilling cycle.

Field Application and future work

Following the continuous upgrading, implementation and optimization of the developed methodology for drilling of horizontal wells, there has been a continuous improvement in drilling performance after four rounds of drilling. Fig 8 shows that a data comparison of the four rounds of completed horizontal wells. The figure indicates that the average drilling cycle was shortened from 119 days to 77 days, and that the average ROP was increased from 4.4 m/h to 7.2 m/h. Despite the progress made in increasing ROP and shortening the drilling cycle, more work is still required in the future to ensure the cost-effective development of this shale oil resource. This work will include adopting PDC and impregnated diamond hybrid drill bit, heavy-impact load ROP enhancement tools, and drilling deeper wells with horizontal sections exceeding 2000 meters to ensure higher EUR and etc.

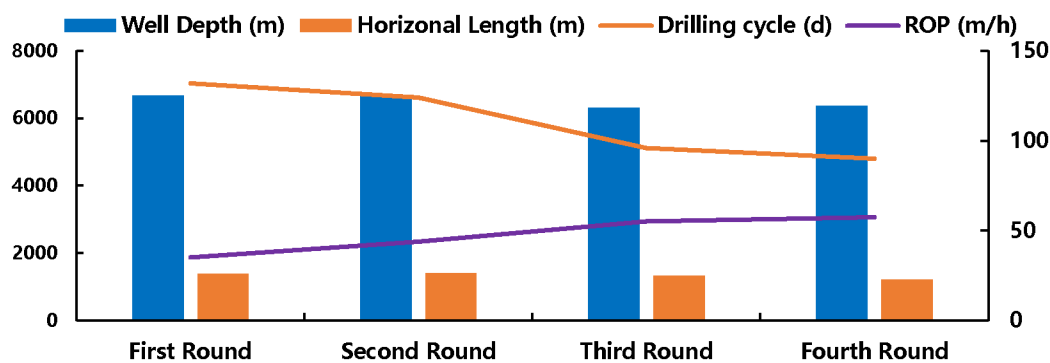


Figure 8—drilling performance after four rounds of drilled horizontal wells

Conclusion

1. The ultra-hard and highly heterogeneous Fengcheng formation posed several challenges to the cost-effective development of this shale oil resource, including low ROP, short drilling footage and a long drilling cycle when drilling the third interval of horizontal wells.
2. The integrated methodology, which includes drilling design optimization based on the three-dimensional prediction of formation drillability, development of customized high-performance drill bits and technical drilling measures for drilling with large WOB, has been developed to overcome the challenges of drilling in this ultra-hard and highly heterogeneous formation.
3. Continuous drilling performance has been achieved by upgrading and implementing this developed methodology over four rounds of horizontal well drilling. The average drilling cycle for completed horizontal wells was reduced from an initial 117 days to 77 days with the average ROP being increased from 4.4 m/h to 7.2 m/h. Further work is planned to shorten the drilling cycle further and ensure its cost-effective development.
4. The challenges, methodology and achievement presented in this paper provide valuable insights into how to achieve optimal performance indicators when drilling in similar ultra-hard and highly heterogeneous formation.

Acknowledgement

The authors would like to thank PetroChina Xinjiang Oilfield Company for the support and permission to publish this paper. Thanks to the support of CNPC scientific and research projects (No.2025ZG58, NO.2024ZZ55).

Nomenclature

CCS–	Confined Compression Strength, MPa
ROP–	Rate of penetration, m/h
WOB–	Weight on Bit, kN
RPM–	Rotation per minute, r/min
RSS–	Rotary Steerable System
HWDP–	Heavy Weight Drill Pipe
PDC–	Polycrystalline Diamond Compact
MWD–	Measurement While Drilling
LWD–	Logging While Drilling
EUR–	Estimated Ultimate Recovery

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